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A THESIS

Submitted to the faculty of the Graduate School of Creighton
University in Partial Fulfillment of the Requirements for the degree of
Master of Science in Medical Physics in the Department of Physics

Omaha, NE
October 15, 2025

Abstract

The abstract is a maximum 350 word, double-spaced summary of your work. The word count limit is important for upload to the Creighton Digital Repository.

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Acknowledgments

Thank your research team, advisor, committee, family and friends, etc. See previous theses for examples.

I would like to thank Kian for introducing me to Latex, Kyle and Bong for troubleshooting help, and Dr. Nichols for his help with thesis creation.

For Dr. Nichols

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Chapter 1

Introduction

Medical diagnostics heavily rely on imaging tools such as Magnetic Resonance Imaging (MRI), a powerful and evolving application capable of showing high contrast sensitivity to soft tissue while not depending on ionizing radiation. However, safety concerns may arise during cardiac MRI procedures, electro cardigram (ECG) leads are generally cataloged as MRI unsafe because of the strong magnetic field and radiofrequency (RF) pulses that may interact with these leads.

More than 300,000 individuals are receiving new cardiac implantable electronic device (CIED) implants every year in the U.S. alone, reported the Heart Rhythm Society (HRS) in 2024 [2]. Some of these patients will eventually require device explantation or revision, necessitating surgery [3]. For this reason, a well established protocol to monitor cardiac rhythm exists, it involves the use of temporary epicardial leads, however there is a recognized practice when these leads are removed (only if significant resistance is encountered) to cut the wire flush with the skin, intentionally leaving a small segment inside the patient to prevent immediate complications [4, 5, 6].

Temporary epicardial pacing leads are frequently utilized as part of postoperative cardiac rhythm control procedures, for example, leads are frequently placed during open-heart surgery to regulate cardiac rhythm postoperatively, then are typically cut

at the skin surface leaving behind a short length of wire embedded in the tissue [7]. When the surgical danger of removing these leads is too high, they are left in place to prevent further complications. These cases introduce long-term repercussions, most notably the loss of access to MRI.

According to ASTM F2503, a device labeled as MRI safe is expected to be non-conductive, non-electrical, non-magnetic, and to present no known hazards in the MRI environment [8]. However, this classification does not apply to all epicardial leads, as they are rarely physically characterized after removal, nor properly labeled by vendors. By the time they are cut, these leads have fulfilled their clinical purpose, and systematic safety evaluations are seldom performed.

Although retained leads are nonfunctional and relatively short, they are frequently classified as MRI unsafe due to concerns over RF heating, magnetically induced translational forces and torques, or inadvertent cardiac stimulation [9, 10, 11]. Effectively preventing patients with CIED or cardiac surgery history, who are most likely to need a MRI scan during their lifetime, to not be able to enter MRI room.

There is still a dearth of focused studies evaluating the precise dangers associated with retained epicardial leads, despite the frequent use of such leads and their repercussions. Existing clinical recommendations tend to be too cautious and frequently disqualify these individuals from MRI eligibility even when there is little evidence to support it. Recent studies regarding this problem have shown both no image quality degradation and, most importantly, no serious safety events such as, device malfunction, myocardial injury or arrhythmias, associated with MRI examinations in patients with CIEDs abandoned leads, including epicardial leads [12, 13]. While these work have evaluated to some extent the resulting consequences of RF-induced heating, mechanical forces and conductivity, detail is needed.

Therefore, the primary safety concerns regarding retained cardiac leads during MRI arise from three main interaction mechanisms. First, the strong magnetic field

and its spatial gradient can exert translational forces and torques on the lead, resulting in mechanical displacement along the direction of the magnetic field. Second, the lead can absorb energy from the MRI's (RF) field, potentially causing localized heating that poses a thermal risk to the patient. Third, the RF field may induce alternating currents within the lead, which can electrically stimulate nearby cardiac tissue, potentially leading to dangerous physiological effects.

Being the second the main topic of this document, translational motion and torque. Which remains physically under-characterized and that most of the research done in this subject shows the importance of translational forces in safety about common hospital objects that could potentially become projectiles while disregarding smaller objects.

In this document an explanation of the current state of research on this matter, the main elements involved in the interaction of temporary epicardial leads and MRI, especially in relation to mechanical interactions like translational force and torque are given. Current research, safety recommendations and methodologies to assess the magnetically induced forces involved between an MRI system and epicardial leads were explored. And the development of a in-lab methodology capable of testing these micro-Newton forces with precision later carried into clinical work was documented. All these, in order to clarify whether the static magnetic field in MRI poses a real risk to patients with retained leads.

To properly situate this work, the following section reviews pertinent literature on MRI devices, MRI safety principles, electromagnetic theory, lead design, lead-MRI interactions and previous studies to establish the technical and clinical context needed to fully understand the motivations and significance of this study.

1.1 Theoretical Background

1.1.1 Magnetic Resonance Imaging

MRI or Magnetic Resonance Imaging is a medical imaging modality that uses magnetic fields and radiofrequency (RF) energy to produce images, rather than ionizing radiation like X-rays making it inherent safe. Moreover, MRI machines are capable of archiving high contrast sensitivity to soft tissue differences because of the reliance on micromagnetic properties of the tissue and its inherent differences leveraging on Proton Density (PD) and the relaxation phenomena.

MRI, as the name suggests, leverages on the magnetic properties of matter to capture information ultimately forming an image. It is important, then, to understand these properties and how they are characterized. In a particle scale these effects are shown in protons and neutrons within the nucleus due to their nuclear spin and charge distribution. Hydrogen (specifically its proton, 1H) is the most utilized element in clinical MRI because it has the largest magnetic moment and is highly abundant in biological tissues, primarily in water and fat.

When placed in a strong static magnetic field (B_0), these protons align in parallel (low-energy state) and antiparallel (high-energy state) directions. A slight majority align in the low-energy, parallel direction, resulting in an observable net magnetic moment for the sample. As illustrated in Figure 1.1, the net magnetic moment of hydrogen protons forms the basis of MRI signal generation.

A common MRI has several components that interact with matter around it, for example, heat and electricity interactions could be a concern on RF fields or Gradient Coils. For mechanical interactions, the interest falls under the question “What are the mechanisms in which the static magnetic field could possibly interact with wire or thin cylinder like devices?”.

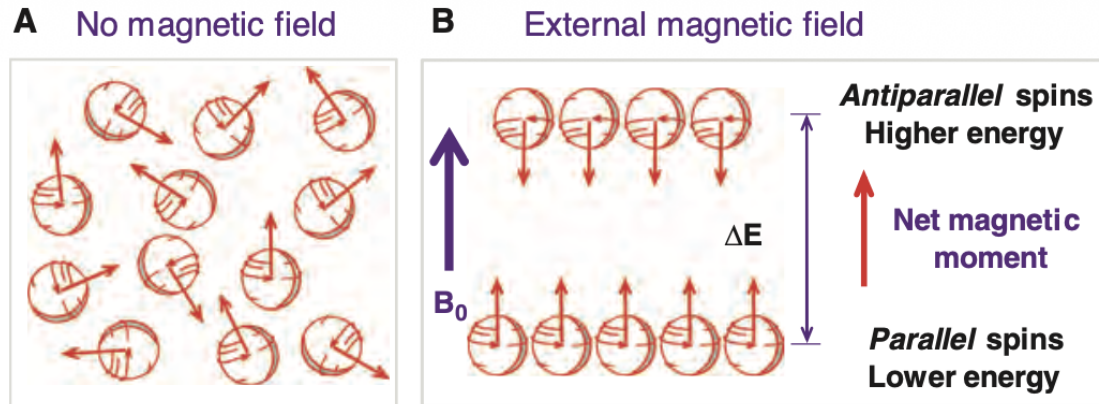


Figure 1.1: Schematic illustration of how hydrogen protons (^1H) behave in the presence of a strong static magnetic field (B_0). A: In the absence of a field, proton magnetic moments are randomly oriented, producing no net magnetization. B: When exposed to B_0 , protons align either parallel or antiparallel to the field. [14].

This subsection provides a review of the physics and its implications for the clinic and the lab setting. Most of its content was based on physics and imaging well known textbooks, including Young's & Freedman's University Physics, Griffith's Introduction to Electrodynamics, and Bushberg's Physics of Medical Imaging [14]

The Static Field

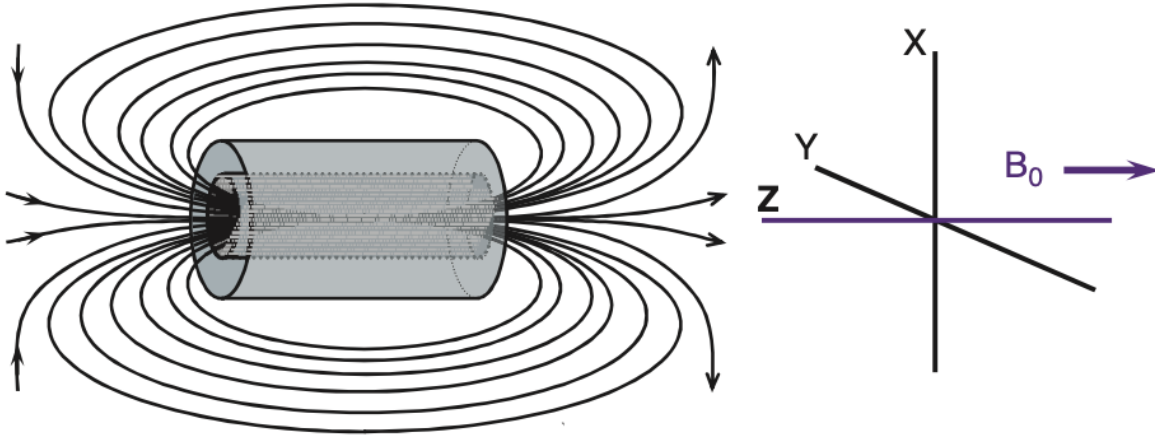
Magnets are at the core of MR systems, and their performance is judged by field strength, temporal stability, and homogeneity. The strength of a magnetic field, also known as magnetic flux density (B), is measured in Tesla (T). An older, alternative unit is the gauss (G), where $1 \text{ T} = 10,000 \text{ G}$. For perspective, Earth's magnetic field is significantly weaker, around 0.00005 T (0.05 mT or 0.5-1.0 G).

Magnetic fields can be induced by moving charges in a wire, where the direction depends on the charge's sign and direction, described by the "right-hand rule". Coiling a current-carrying wire, creating a solenoid or electromagnet, superimposes these magnetic fields, significantly augmenting the field strength inside the coil and causing

a rapid falloff outside. The amplitude of the current and the number of coil turns determine the overall magnitude of the magnetic field strength as follows:

$$B = \mu_0 nI \quad (1.1)$$

Where B is the magnetic field, μ_0 is the vacuum permeability, n is the number of turns and I is the current. Common MRI are typically made of superconducting wire-wrapped cylinders. These wires are made from niobium-titanium alloys and show no resistance to electric current when maintained at extremely low temperatures. The wire-wrapped cylinder produce a magnetic field parallel to the cylinder's long axis, usually horizontal (along the patient's cranial-caudal axis in clinical systems).



Let us then follow the physics of a static magnetic field and its interaction with matter with electromagnetism theory, any material placed in a magnetic field develops a magnetization, $\mathbf{M}$, defined as the net magnetic dipole moment per unit volume:

$$\mathbf{M} \equiv \frac{\boldsymbol{\mu}}{V} \quad (1.2)$$

where $\boldsymbol{\mu}$ is the sum of the microscopic magnetic dipoles within the volume V [15]. In practice, the microscopic magnetic field inside a material fluctuates rapidly due to atomic and electronic motion. To describe macroscopic behavior, we use an average field over many atoms, and $\mathbf{M}$ represents this “smoothed” magnetization.

Different materials respond differently to an applied field. Diamagnetic materials, such as water, calcium, and most organic compounds, develop a magnetization opposite to the applied B-field due to paired electrons. Paramagnetic substances, like molecular oxygen, deoxyhemoglobin, or gadolinium-based MRI contrast agents, align weakly with the field, enhancing the local magnetic flux. The degree to which a material becomes magnetized is quantified by its magnetic susceptibility, χ , which is typically small in biological tissues ($|\Delta\chi| < 0.01$). Ferromagnetic materials, by contrast, exhibit strong self-sustained magnetization but are generally avoided in MRI due to safety hazards.

To simplify the analysis of materials in magnetic fields, physicists define the auxiliary magnetic field $\mathbf{H}$:

$$\mathbf{H} \equiv \frac{1}{\mu_0} \mathbf{B} - \mathbf{M} \quad (1.3)$$

where $\mathbf{B}$ is the macroscopic magnetic flux density and μ_0 is the permeability of free space. $\mathbf{H}$ accounts for the part of the magnetic field produced by free currents, separating it from the bound currents induced by magnetization. This distinction becomes useful when analyzing mechanical effects on magnetic dipoles.

Magnetic dipoles interact mechanically with a magnetic field through forces and torques. A dipole $\boldsymbol{\mu}$ in an inhomogeneous field experiences a force:

$$\mathbf{F} = (\boldsymbol{\mu} \cdot \nabla) \mathbf{B} \quad (1.4)$$

and a torque:

$$\boldsymbol{\tau} = \boldsymbol{\mu} \times \mathbf{B} \quad (1.5)$$

From these two equations we can solve for more specific physical setups. For instance, consider a pendulum that is exposed to a strong

In general, forces are described as a function of the static field strength (B_0), its spatial gradient (∇B_0), and the magnetic susceptibility (χ) and density (ρ) of the material. A well known and expected behavior is the magnetic pull on high susceptibility materials toward regions of higher field strength, like near the bore entrance of the scanner where the spatial gradient is highest [4, 14, 16]. For instance, a nonmagnetic stainless steel wire with $\chi \approx 103$ ppm may experience a force equivalent to $\sim 30\%$ of its weight, whereas ferromagnetic materials like pure nickel can experience forces exceeding 20,000 times their own weight. That is more than enough to become dangerous projectiles unless restrained [4, 16].

Gradient coils and RF coils?

Do I need to explain other components of the MRI? Gradient coils and RF coils, are relevant in the broader view like from brights thesis. and future work.

Once RF is understood we can explain how this affect the leads resulting in heating and electric conductivity, based on the past local research and literature. (Bright's thesis)

Safety

the safety the 5 gauss line the examples of safety ASTM translational and torques studies the current protocols to assess objects near MRI field how this is different for us, size and suceptibility

ASTM F2052-21 offers a standardized technique for assessing magnetically induced displacement force on medical equipment in order to set safety standards for such situations. This standard states that a device is MR Conditional if the deflection force it encounters in the MRI environment is less than the force of gravity upon it. This is usually verified if the deflection angle from vertical is less than 45° [17, 18].

1.1.2 Epicardial Leads

CIED systems refer to cardiac implantable electronic devices, including pacemakers, implantable cardioverter-defibrillators (ICD)s, cardiac resynchronization therapy (CRT) devices, and implanted rhythm monitors. These are advanced medical devices used to monitor and regulate heart rhythm disorders and leads are a component of CIED systems.

In practice, leads are a type of very thin, ~ 4 to 5 F (French gauge, meaning 1.3 to 1.7 mm), electrical wire that connects a pulse generator to the heart. Figure 1.2 shows the two types of leads that are used, epicardial leads; which are implanted directly on the epicardium (the outer surface of the heart) and Transvenous leads, inserted through the veins and positioned inside the heart's chambers, making contact with the endocardium (the inner lining of the heart). All these are prone to complications like fracture, infection, venous obstruction (transvenous only), and require complex extraction procedures [19, 20].

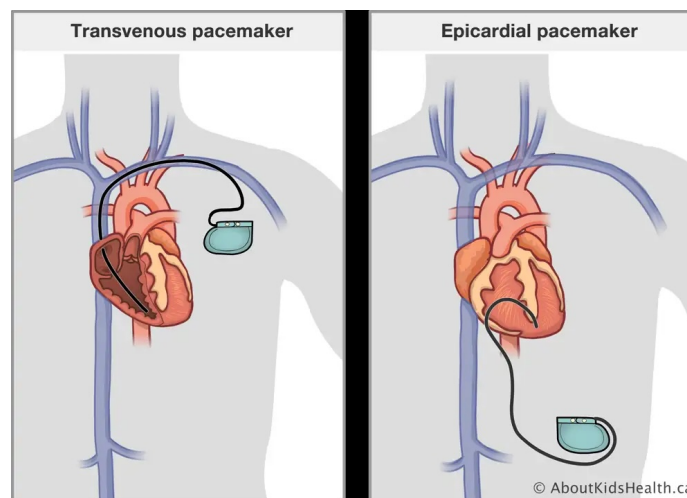
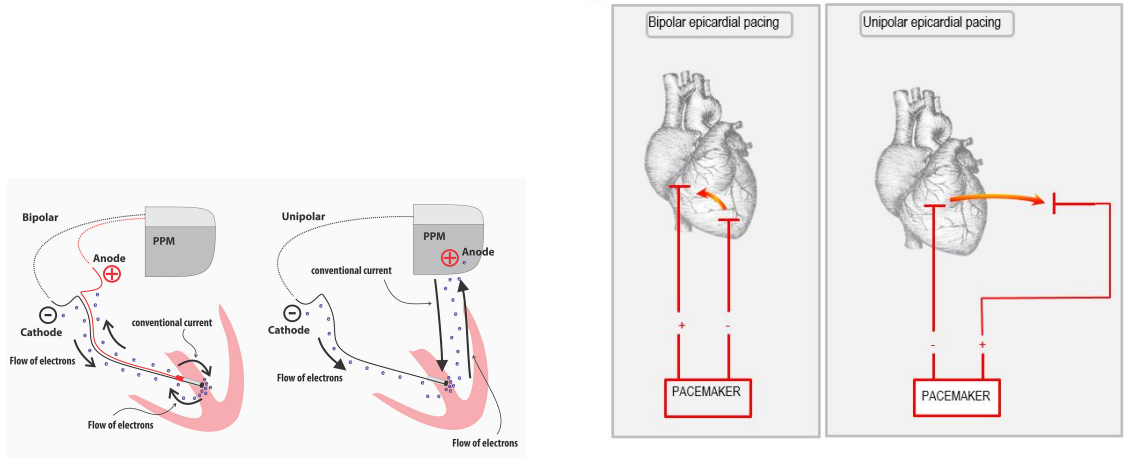


Figure 1.2: Comparison of transvenous vs epicardial lead placement in pacemakers (adapted from [1]).

The position of leads is an important distinction. Clinically, these leads are indicated for different populations and diagnostics therefore carry a variety of man-

agement decisions (leaving vs extracting) and different complications (percutaneous lead extraction or transvenous lead extraction). [13, 21] Physically, different contact with tissue means mechanic interaction, an example is that transvenous leads may be cooled by blood affecting its overall heating capacity. Or that any MRI-induced displacement force or vibration translates directly to mechanical stress for epicardial leads, being sutured directly to the epicardial tissue. While tension on transvenous ones may be somewhat dampened by surrounding blood and vessel walls.

Within epicardial leads, design may vary, examples are from lead polarity and body structure. On unipolar leads, a single conductor and one electrode (the cathode) are in contact with the heart, while bipolar leads contain two conductors that connect to two electrodes on the lead itself, a tip electrode (cathode) and a ring electrode (anode). In figure 1.3 the comparison of these types is shown, on the left side (1.3(a)) depicts the current flow for each case while the right side (1.3(b)) shows from outside the position of the anode relative to the heart.



(a) Unipolar vs. bipolar epicardial lead configuration.

(b) Relative electrode positioning of unipolar and bipolar epicardial leads.

Figure 1.3: Comparison of unipolar and bipolar epicardial pacing leads. (a) Illustrates the distinction in current flow pathways using a schematic of the heart, adapted from [22, 23]. (b) Depicts the relative anode and cathode positions in unipolar and bipolar epicardial leads, adapted from [24, 23].

Bipolar leads then, have a body structure design variation. As shown in Fig. 1.4, coaxial design features an inner coil conductor nested within an outer coil conductor, with each wrapped in a separate layer of insulation while coradial design consists of a single coil containing two parallel, intertwined conductor strands, each individually insulated [23]. Furthermore, epicardial leads, when abandoned, encompass this diversity in design and composition as all components of the leads have different materials.

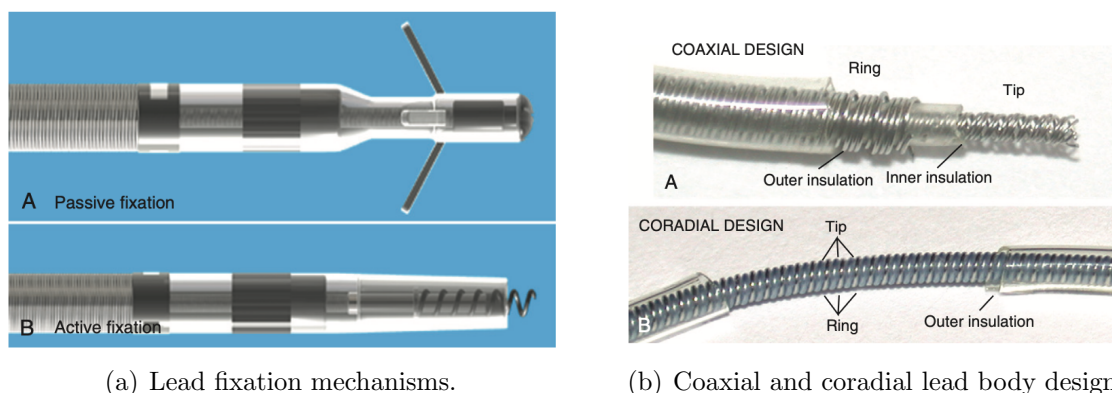


Figure 1.4: Lead fixation and body design variations in pacing leads. (a) Passive-fixation leads rely on flexible tines that anchor within myocardial trabeculae, while active-fixation leads employ a retractable helix that screws into the myocardium. (b) Coaxial designs use two nested coil conductors with separate insulation layers, whereas coradial designs incorporate parallel, intertwined conductors within a shared lumen. Adapted from [23].

The several components of an epicardial lead are made of different materials chosen for their biocompatibility, durability, and electrical properties. Electrodes are made from inert materials to minimize corrosion and inflammatory reactions over the long term. These include titanium, platinum-iridium alloys, Elgiloy (an alloy of cobalt, iron, chromium, and other metals), and vitreous carbon. Similarly, conductors which carry the electrical signals between the pulse generator and the electrode are commonly made out of MP-35N (alloy of nickel, cobalt, chromium, and molybdenum.) Then, insulation, a protective outer coating of the lead that prevents short-circuiting

made of different polymers like Silicone Rubber (Polysiloxane), Polyurethane, Fluoropolymers.

Finally, epicardial leads are affixed directly to the heart’s outer surface during a surgical procedure in mainly two ways, as shown in figure 1.4(b). Suturing and Screwing, meaning the electrode is designed to be sutured directly onto the epicardium. While screwing refers to an active-fixation helix (screw), typically made of a platinum-iridium alloy or a similar metal, screwed directly into the myocardium [23]

From figure 1.4 one can see some of the challenges when retracting the epicardial lead, for this practical reason they are commonly abandoned after its purpose has been fulfilled. An abandoned epicardial lead could measure full-length or fragments after being left behind. Some fragments are described as long as 5cm, being 2cm an average length. Further, longer leads are reported up to 30 cm or couple fragments adding to about 21 cm. In diameter size may be compared to standard transvenous pacing leads, that of about half a millimeter to 1.5 mm. And are typically narrower than standard transvenous leads ($\sim 5F$, $\simeq 1.7$ mm outer diameter), often measuring between 0.5 mm and 1.5 mm.”

The physical length of the wire can impact the resulting interactions with the magnetic field, as more of the material in the human body more of it can interact. Table 1.1 summarizes the materials found on a epicardial lead and relates to its magnetic susceptibility.

Table 1.1: Qualitative magnetic susceptibility of common epicardial lead materials and their relevance for MRI safety. [8, 25, 26, 27]

Component	Material	Magnetic susceptibility	Relevance for MRI safety
Electrode	Titanium	Paramagnetic	Low; minimal torque/force in MRI
Electrode	Platinum–Iridium alloy	Paramagnetic	Low; generally MRI safe
Electrode	Elgiloy (Co–Cr–Fe–Ni alloy)	Paramagnetic	Moderate; may experience slight translational force
Electrode	Vitreous carbon	Diamagnetic	Minimal; very low MRI interaction
Conductor	MP-35N	Weakly paramagnetic	Low; minimal contribution to MRI forces
Insulation	Silicone rubber (polysiloxane)	Diamagnetic	Negligible; non-conductive and non-magnetic
Insulation	Polyurethane	Diamagnetic	Negligible; non-magnetic
Insulation	Fluoropolymer (PTFE)	Diamagnetic	Negligible; MRI inert

Current institutional recommendations regarding MRI in patients with retained epicardial leads remain conservative despite evolving evidence. Major guideline bodies, including the Heart Rhythm Society (HRS), the European Society of Cardiology (ESC), and the Centers for Medicare & Medicaid Services (CMS), allow MRI in patients with MRI-conditional or certain legacy cardiac implantable electronic devices (CIEDs), but continue to restrict or discourage imaging in patients with abandoned or epicardial leads.

Specifically, at the time of the HRS 2017 consensus and the 2021 ESC pacing guidelines noted that there was insufficient data to comment on the safety of MRI with an abandoned cardiac lead. And that if a patient had an MRI-conditional system along with an abandoned lead, lead extender, lead remnants, fractured lead(s), or surgically implanted epicardial leads, the system was considered non-MRI-conditional. Leading to a Case-by-Case consideration for whether to image or not and in which modality. CMS similarly excludes abandoned or epicardial leads from coverage due to perceived insufficient safety data, even though large observational studies indicate that modern epicardial leads (implanted after 2000) may undergo MRI safely under carefully monitored conditions [12, 28].

This cautious stance of institution remark the need for experimental characterization of mechanical interactions between retained epicardial leads and MRI fields. Thus, providing a rationale for the present study. By quantifying these forces under controlled laboratory conditions, the work shown in this document directly addresses whether current institutional restrictions are justified and lays the groundwork for evidence-based updates to clinical safety policies.

1.2 Literature Review

ASTM-F standards and mechanical risks [29] safety [30] Contraindications

both [17],

force [31]

torque [32]

Do I need to add a case when a metal object was attracted violently towards the bore? Just to make an impact...

Previous studies made in our institution seek to illustrate the interaction of leads with the RF field in terms of energy absorbed and heat produced. It showed that lead length, type, and orientation significantly influence heating behavior. This study found that short leads (< 13 cm) posed no significant thermal hazard during MRI at a specific absorption rate (SAR) of 2 W/kg and is thoughtfully described as Master Thesis in 2021 [33, 27, 4].

Along the same line of thought, we aim to cover the mechanical aspect of lead-MRI interaction covering (1) Characterization of the static magnetic field, (2) quantification of translational forces near the bore entrance, (3) Torque evaluation from bore entrance to isocenter.

While some preliminary work has been conducted in our lab on these mechanical concerns, it remains unpublished, making this thesis a critical step in formalizing and extending that foundational effort.

1.3 Scope of Research

The primary objective of this study is to experimentally characterize torque and translational magnetic forces acting on retained epicardial pacing leads near and inside the MRI bore, and to map how those forces evolve with distance from the isocenter.

The wider research question is: Do retained epicardial pacing leads experience torque or translational magnetic forces within an MRI environment that are clinically significant enough to justify current safety restrictions? A particular research question

then follows: How to implement current methodologies to measure micro-Newtons on a clinic environment?

This aims to directly assess whether the magnitude of such forces constitutes a meaningful mechanical risk in clinical settings. A lab methodology was first developed, working towards clinical environment and rigor addressing potential issues. With it we aim to provide experimental evidence and addressing a gap in the literature.

In the lab, investigation used a Polyvinyl Chloride (PVC) pendulum-based measurement setup similar to ASTM F2052 [18] for the displacement angle and a custom-made solenoid simulating a magnetic field similar to that found at the MRI bore entrance. Displacement in millimeters of a metallic rod exposed to the field were recorded to ultimately estimate the translational force.

A second setup for torque measurements was developed. A set of neodymium magnets would create the static field while a metallic rod would be fixed in a position perpendicular to the field with an attachment and be allowed to rotate, counter forces would then be measured out the attachment with a circuit and sensor of our own fabrication. A second iteration of this setup would include a Helmholtz coil to test on field variation.

These methods allow for high sensitivity in detecting small magnetic forces in a controlled environment, without the complications of patient variability or direct MRI scanner use. The pendulum approach provides a clear physical model that isolates translational force components for precise characterization. The torque model allows to independently assess strength of force exerted.

In the following chapter, a thorough description of the experimental setup will be given. This includes the caveats found and their respective solutions with detailed documentation on the artifacts and tools created for the investigation.

The thesis consistently engages with 1 of the 3 interaction modes of concern between MRI and retained leads: mechanical forces, RF-induced heating, and induced currents. In these mechanical forces aspect we particularly care about translational and rotational displacement due to static field gradients. Concepts from electromagnetism, MRI safety standards, and biomedical device regulation form the thematic backbone of the work.

Measurements are interpreted in the context of magnetic susceptibility theory and compared to relevant thresholds from the literature [33, 18]. These techniques will help with more thorough evaluations of MRI safety in patients with retained temporary epicardial leads and set the stage for a future clinical adaptation of the protocol employing a 3T MRI scanner.

Chapter 2

Materials and Methods

The objective of this study was to measure and quantify MRI-induced translational and torque forces experienced by an post-surgical epicardial pacing leads when exposed to the static magnetic field. Safety limits for MRI magnetic fields are well established and described in the Introduction, based on ASTM F-series guidelines. Following these methods, we constructed non-magnetic test apparatus specifically designed to measure these forces with the precision required for clinical applications.

However, clinical access for experimental testing posed a significant challenge, as MRI scanner time is typically associated with long scheduling delays and short available measurement periods. Therefore, a secondary objective of this work was to improve the quality of our experimental setup in order to maximize the efficiency of data acquisition during these limited observation windows. In this chapter, a layout of the experiments conducted is documented.

Previous students developed the laboratory-scale instruments and associated software necessary to evaluate magnetic forces in a controlled environment. This was achieved by generating a smaller but reproducible magnetic field in the laboratory, enabling characterization of materials with relatively high magnetic susceptibility. While this laboratory-generated field was not sufficient to directly measure or char-

acterize the epicardial lead itself, it allowed for evaluation of the accuracy and measurement capacity of each instrument, as well as for the observation of the physical phenomena described above. Furthermore, given the proportional nature of the magnetic effects on a material, it was expected that, when tested in the much stronger (3 T) magnetic field of a clinical MRI system, the small magnetic susceptibility of the epicardial lead would not present a significant limitation in terms of measurement accuracy.

Three distinct experimental setups were involved in this study: (1) a spatial field mapping platform to characterize B_0 and ∇B_0 , (2) a pendulum-based deflection apparatus to assess translational force through angular displacement, and (3) a torque measurement system for rotational force assessments. In this chapter we will explain our setup and rationale, as well as the proven modifications to the instruments in order to provide a reproducible map of action to test our instruments.

2.1 Magnetic Field Generation and mapping

Magnetic fields were generated in multiple ways, seeking to archive the most reliable or similar situation as one would encounter on a clinic. The most intuitive way to accomplish this in a lab setting was to use neodymium magnets with the only problem being static or step-wise change of its magnetic flux, but proving to be quite effective and strong.

Another approach included using a custom-built air-core solenoid, approximately 15 cm in length and 10 cm in outer diameter. The solenoid was densely wound with copper wire (~ 1 mm diameter), forming an inner core of approximately 5 cm. A Pasco Scientific SF-9584 DC power supply was used to deliver currents in the range of 0–4 A, corresponding to magnetic field strengths from 1 to 18.9 Gauss for this solenoid. This magnet was particularly useful for magnetic field and translational

forces measurement tests and was operated in a continuous mode; no rest periods were applied between trials due to the time required to stabilize the pendulum after each adjustment.

Finally, in line to mimic that of a static but increasingly stronger magnetic field inside an MRI we used a set of educational Helmholtz coils

Further describe this coils, size, number of turns, pasco, series so on

Around our torque contraption allowing us to see changes with its strength in real time using the same Pasco Scientific SF-9584 DC power supply.

2.1.1 Magnetic Field and Gradient Mapping Platform

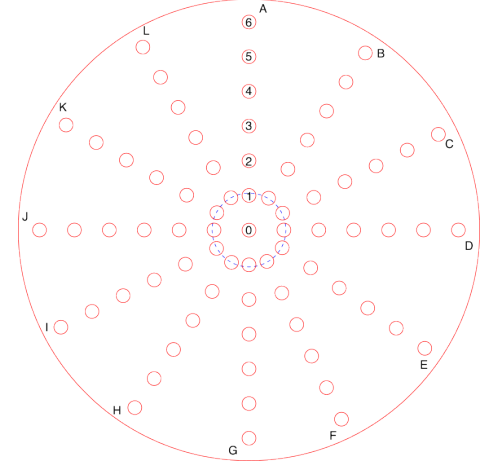
Magnetic field strength (B_0) and gradient (∇B_0) measurements were performed using a commercial Gaussmeter (Model GM2, AlphaLab Inc., Salt Lake City, UT). The GM2 operates on the Hall effect principle, wherein a voltage is generated across a probe when exposed to a perpendicular magnetic field. This voltage is proportional to the local magnetic flux density, enabling precise spot measurements of field magnitude. The Gaussmeter used has an accuracy of 1% of the DC reading in the 16°C to 29°C range.

To ensure spatial accuracy and reproducibility, a fixture made from rigid PVC was constructed by previous Creighton University students [33]. The platform comprises a 300 mm square base and a 280 mm diameter vertical circular plate, perforated with a 5x5 grid of probe-access holes spaced at 20 mm intervals as shown in figure 2.1. Each access point was numerically indexed to facilitate 2D field mapping in the X-Y plane.

Once the field-mapping phantom and Gaussmeter probe are co-aligned, data capture was streamlined using LabVIEW to be sufficiently flexible and precise for our proximity measurements near an MRI. The manufacturer’s default software, AlphaApp, provides basic data-logging, recording, live screen streaming, and plotting of



(a) PVC-based mapping platform.



(b) Example field map output from MRI fringe region.

Figure 2.1: PVC-based magnetic field mapping platform. Used for spatial B_0 and ∇B_0 measurement. As shown in Ferreira 2017 [33, 34].

measurements from USB-enabled AlphaLab meters[35]. However, AlphaApp does not expose control over sampling frequency or allow real-time adjustment of acquisition parameter, limiting fine temporal resolution when the probe is moved into steep field gradients. In contrast, using LabVIEW Virtual Instrument (VI) (VI) interface allows for configurable sampling intervals, buffer control, and precise timing protocols, enabling us to push the device for much higher accuracy as the probe approaches and crosses isocontour regions within the MRI fringe. We developed that VI by leveraging the AlphaLab Data Acquisition Communication Protocol to command the device directly via USB, providing control over timing and sample rate [36].

The data acquisition procedure for magnetic flux mapping is organized in this VI, whose basic logic is summarized in Figure 2.2. Upon startup, the VI initializes serial communication with the Gaussmeter, loads user-defined parameters (such as file name, mode of acquisition, delay time, and probe coordinates), and clears buffers in preparation for logging. The user may select either a continuous or an individual sampling mode. In continuous mode, the VI iterates through a defined number of data points with a fixed delay between samples; in individual mode, each data capture

is triggered manually, allowing for precise positioning or timing. For both modes, each cycle involves reading the spatial coordinates, sending a measurement command, recording the returned magnetic flux value, and updating a graphical display as well as the output file.

While the flowchart reflects the core logic, it does not fully convey the customization added to the VI interface. For example, the front panel includes input fields for patient table (couch) parameters such as couch speed and displacement points. As mentioned above, phantom is discretized with radial and angular indices: angular positions span 12 sectors clockwise from the 12 o'clock position (0 to 11 or A to L), and radial depths range from the center (0) to the outer edge (6). The probe's spatial location is represented in polar coordinates, and the system plots magnetic flux as a function of position rather than time, allowing for intuitive visualization of the MRI fringe field across the phantom surface.

In clinical use, the entire fixture is intended to be shifted axially along the Z-direction of a horizontal bore MRI scanner using controlled couch increments (e.g., 10 cm). This translation allows for volumetric sampling of B_0 over several planes. The spatial magnetic field gradient ∇B_0 can then be estimated numerically using finite difference calculations between adjacent measurement points. This method has previously been applied to MRI mapping in the work by Ferreira [34].

2.2 Translational Force Measurement via Angular Deflection

2.2.1 Measurement Device

To assess MRI-induced translational forces in a laboratory environment, Creighton University developed a pendulum-style apparatus following the geometry and physics

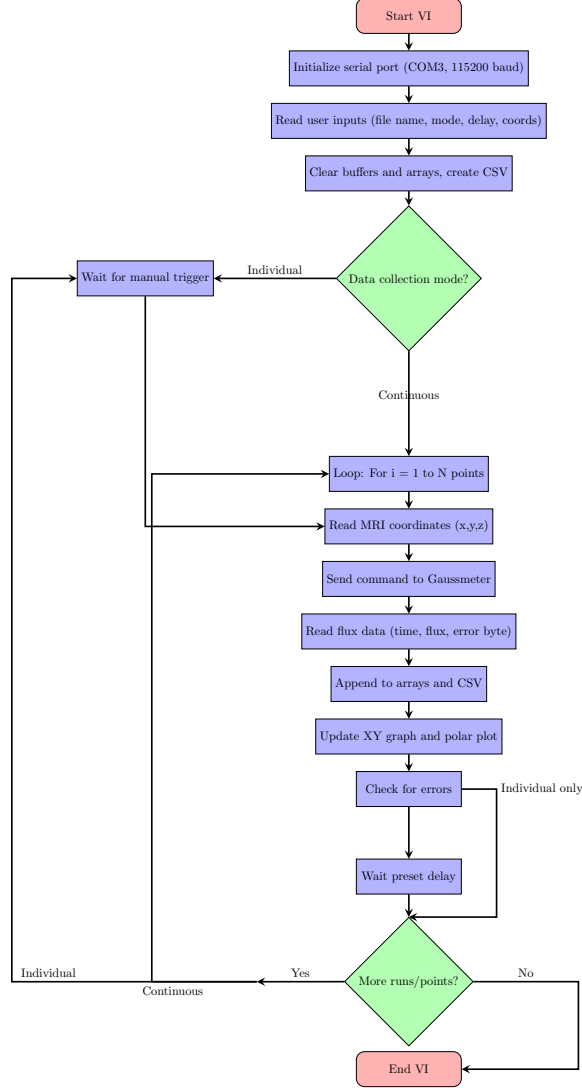


Figure 2.2: Flowchart of the Gaussmeter VI data acquisition algorithm.

outlined in ASTM F2052-15[18]. The vertical post includes a screw clamp mechanism that enables height adjustment; however, a default suspension height of $L = 1$ meter was used for consistency across trials.

The experimental frame was constructed entirely from PVC components, including a vertical post of about 1.5 meters and a rigid base for stabilization. Attached to the pole a set of rings were attached to the pole with a nylon screw to held positions. The lower one features a slit, where a custom made Kydex ruler sits held by smaller nylon screws. This ruler is interchangeable. The upper ring shows a rubber clamp

from which a string passes. Then the length of suspension, $L = 1$ meter, was set as the distance between these two rings, specifically from the last contact point of the string to the rubber on the top ring to the sample hanging free at ruler's height as shown in Figure 2.3.

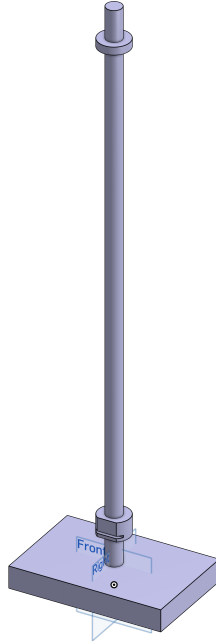


Figure 2.3: CAD render of the pendulum apparatus created using CATIA.

The sample was a solid ferromagnetic cylinder of length = 6 mm, diameter = 3 mm, and mass = 0.3648 g therefore density $\rho = 8.6 \text{ g/cm}^3$). It was suspended by a 955 mm-long sewing thread, tied with a simple knot at its center. For clinical applications, a 0.25 mm nylon monofilament is recommended. A monofilament nylon thread has low stretch, complies with MR-safe properties (Non-metallic, Non-conductive, Non-magnetic), and reduces variability introduced by torsional drift or thread elasticity [17, 18]. The bottom of the sample was free to swing, forming a pendulum under gravity. A horizontal ruler, fixed along the Z-axis, served as the visual reference to quantify lateral displacements.

Lead Holder Frame Design

To accommodate epicardial lead positioning while maintaining MRI compatibility, we propose the use of a lightweight, non-magnetic bee comb mesh fabricated from Polylactic Acid (PLA). The suggested mesh dimensions are 16 mm length by 14 mm wide, with a thickness of 1 mm, for a total volume of less than 100mm^3 equivalent to around a tenth of 1 gram given the 1.24 grams per cubic centimeter (g/cm^3) PLA density. As shown in figure 2.4.

The mesh needs to be suspended such that the lead axis was aligned approximately parallel to the magnetic field gradient (Z-axis). The mesh's lightweight design aimed to minimize its contribution to the total magnetic force, however its mass was not explicitly subtracted from the translational force calculations.

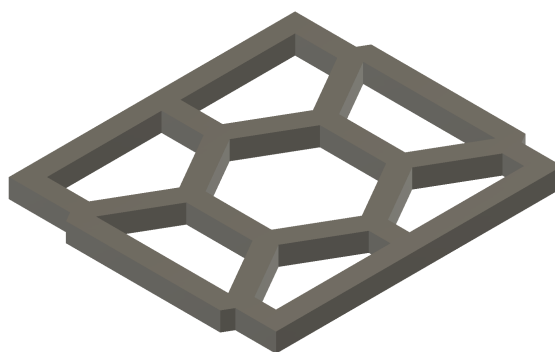


Figure 2.4: The design on lead holder comprises a rectangular lightweight bee comb mesh to wrap the epicardial lead around and hold it in a pendulum with a nylon string

A summary schematic of the apparatus and frame are included in Figures 2.3 and 2.4, with additional engineering drawings available in Appendix A

In Lab Measurement Protocol

Magnetic field measurements were performed using the **AlphaLab Inc. Gauss-meter** described before. The setup consisted of a custom-made solenoid connected to a Pasco power supply capable to deliver up 8 A acting as variable magnetic field

generator for our experiment, the subject - a small ferromagnetic rod - was attached to a string pending from the MRI safe pole and angle of deflection was measured visually with a ruler on the subject horizontal displacement towards the coil. The Gaussmeter was positioned such that it allowed for small displacements and magnetic field measurements at the same time as angle deflection.

Figure 2.5 shows the experimental lab setup used to assess the methodology accuracy and reliability with all used components.

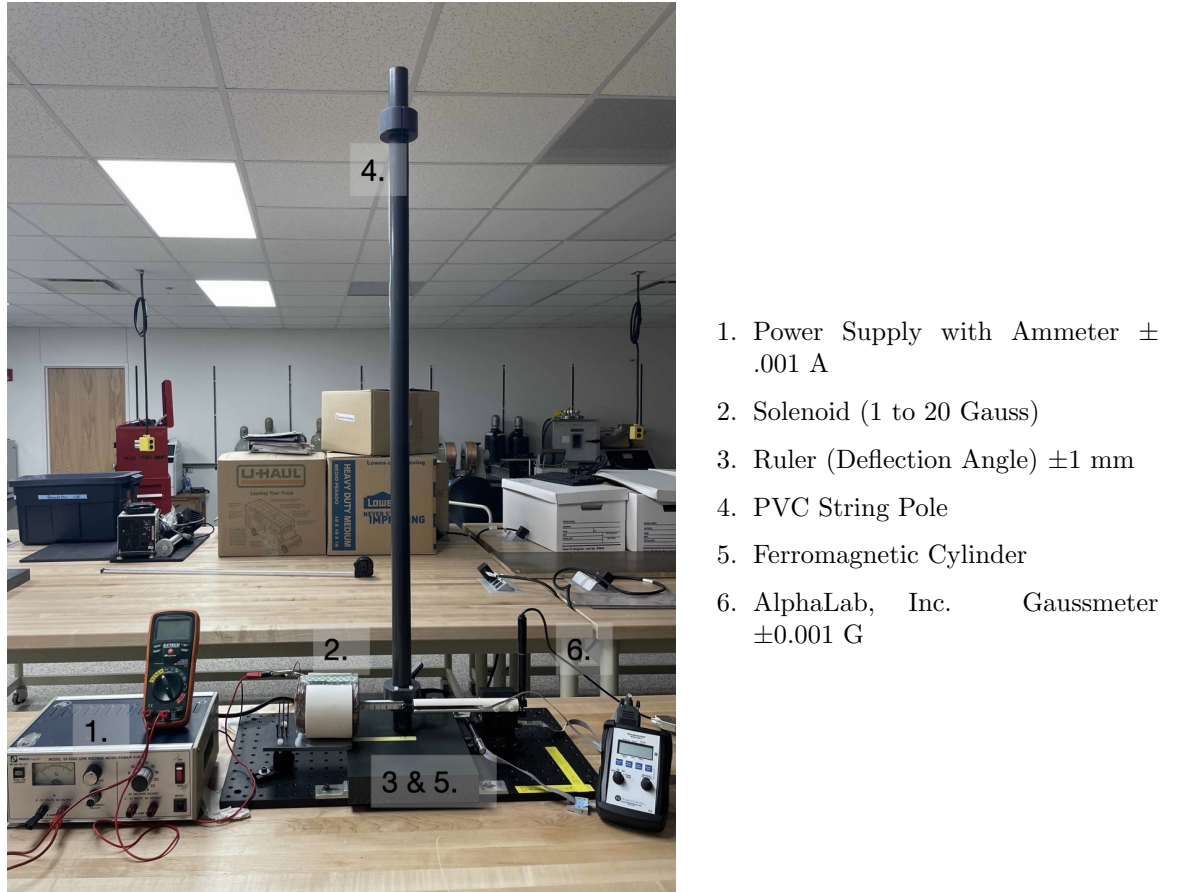


Figure 2.5: In-lab experimental setup for MRI translational force assessment. Labeled components include the solenoid, pendulum fixture, measurement tools, and control equipment.

To characterize local field gradients, measurements were taken at three positions: at the pendulum's equilibrium position, 0.5 mm below, and 0.5 mm above, yielding values of $B_{-0.5}$, B_0 , and $B_{+0.5}$, respectively. Each position was sampled **nine times**

per current level, and gradients were computed using python pandas library and the central finite difference method:

$$\frac{dB}{dz} \approx \frac{B_{+0.5} - B_{-0.5}}{1 \text{ mm}} \quad (2.1)$$

Once the pendulum reached equilibrium under each magnetic condition, the horizontal displacement on z was recorded visually using the ruler as a reference. With known suspension length L , the **deflection angle** α was calculated as:

$$\alpha = \tan^{-1} \left(\frac{x}{L} \right) \quad (2.2)$$

Following ASTM F2052-15, the translational force exerted on a device due to a static magnetic field spatial gradient can be evaluated through angular deflection. In this setup, the test object is suspended in equilibrium, and the balance of forces leads to:

$$\tan \alpha = \frac{F_m}{mg} \quad (2.3)$$

Where α is the deflection angle from vertical, F_m is the magnetically induced translational force, m is the mass of the object, and g is the gravitational acceleration. Rearranging gives:

$$F_m = mg \tan \alpha \quad (2.4)$$

This is also derived from Newtonian force balancing as in Appendix X2 of ASTM F2052-15 (Fig 2.6), and is valid even in cases when the magnetic force does not operate exactly at the center of mass. As a result, the approach is reliable for evaluating a variety of devices.

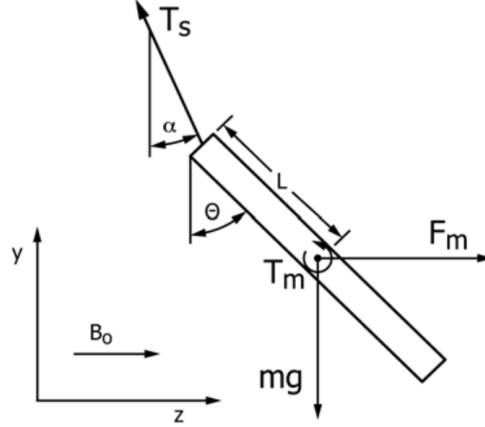


FIG. X2.1 Free Body Diagram of Device in Magnetic Field

Figure 2.6: Free-body diagram of device in magnetic field (adapted from ASTM F2052-15, Appendix X2).

For materials where magnetization is induced by a static field (e.g., paramagnetic or unsaturated ferromagnetic materials), the translational force can also be expressed in terms of volumetric magnetic susceptibility χ , assuming linear response:

$$F_m = \frac{\chi V}{\mu_0} B_0 \frac{dB_0}{dz} \Rightarrow \chi = \frac{\mu_0 F_m}{V B_0 \frac{dB_0}{dz}} \quad (2.5)$$

Combining this with the earlier result:

$$\chi = \frac{\mu_0 mg \tan \alpha}{V B_0 \frac{dB_0}{dz}} \quad (2.6)$$

This provides a direct estimation of magnetic susceptibility from angular deflection measurements, provided the field and its gradient are known.

The ratio is expressed as follows using a different, dimensionally reduced form from ASTM F2052-15's Appendix X3:

$$\frac{\chi}{\rho \mu_0 g} = \frac{\tan \alpha}{B_0 \frac{dB_0}{dz}} \quad (2.7)$$

This formulation explains why the measured deflection angle α (and thus F_m) depends not on χ or B_0 independently, but on their product. Practically, this insight justifies our lab methodology: although our solenoid generates only 20 Gauss, we used a highly ferromagnetic sample (large χ), allowing us to replicate clinically relevant force-to-weight ratios (F_m/mg).

In the clinical MRI environment, B_0 may reach 3 T, but the χ of implanted materials is often orders of magnitude smaller. This inverse balance preserves the translational force regime, and supports translating results from lab simulations to clinical conditions.

All measurements were logged in **Microsoft Excel**, and photos/videos were captured via smartphone to document behavior. A minimum of **three trials** were performed for each current level; however, only the final (most consistent) dataset was retained for analysis. Data were exported to a Jupyter Notebook for clean up and processing with pandas. Then plotted to evaluate $\tan \alpha$ vs. field gradient force. Propagated uncertainties in α and dB_0/dz were computed using standard error propagation formulas.

Smallest measurable force

The smallest measurable force can be obtained with geometry, as the pendulum forms a right-angled triangle where the vertical line is the pole itself " L ", the horizontal is the ruler " x " and the hypotenuse is the pendulum's string as depicted on figure 2.5. Both of these measurements have a 1mm accuracy and equips us with a 1m by 1mm triangle

Max and min forces measured

In Clinic Measurement Protocol

To guarantee safety and repeatability, the measurement of MRI-induced translational forces on the test items was carried out in accordance with a structured methodology. Initially, all personal belongings were removed to prevent safety issues. The apparatus, a PVC pole equipped with a non-magnetic ruler aligned with the pendulum string, was positioned at a precise distance from the MRI bore. Two sets of measurements were performed sequentially: first using a magnetically susceptible test object, and then with the epicardial lead attached, suspended from a light basket and a heavy basket - respectively- to assess force variations. Moving closer to the bore, a step-wise method was followed to measure displacement at certain translational points. Each measurement phase was completed as per the flowchart-guided process, with detailed data recorded in the MEASUREMENT WORKSHEET. The comprehensive protocol, including step-by-step instructions and safety considerations, is provided in Appendix

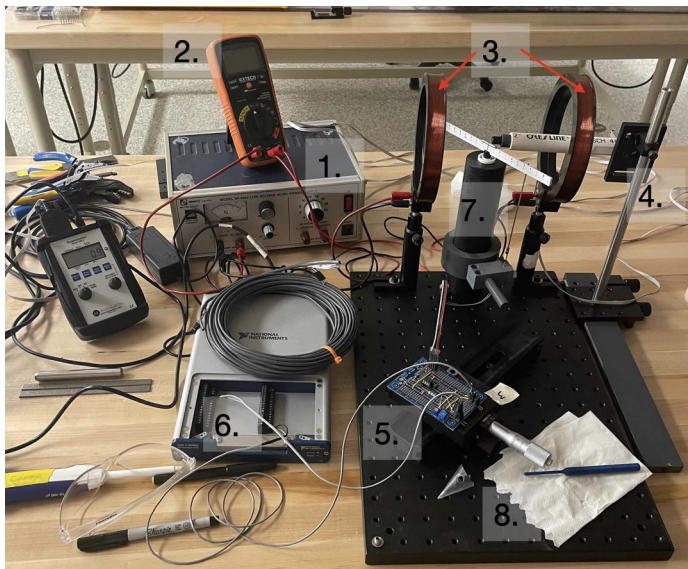
2.3 Torque Measurement via Counter-torque

2.3.1 Torque Measurement Apparatus

On the final part of the assessment, torque on a horizontally oriented subject, as described in the figure ref was measured. To archive this a non-magnetic pole was fabricated with clinical considerations in mind. The pole was made from PLASTIC and is 200 mm tall with 51 mm of diameter it can be attached to a PVC base for support with a nylon screw and it features a space for a ceramic bearing to sit. This ceramic bearing is meant to reduce as much friction for the rotation as possible while keeping everything non magnetic, the bearing has a plastic notch capable of holding a 10 mm thick ruler or rod to spin. This set up is shown in Figure ref . The

induced torque on the subject, from the torque explained in chapter /refch2, would be transmitted mechanically to a non-magnetic ruler which then would apply a force on a custom build force sensor (Gauge sensor). This force sensor would bend on and the counter-force would be measured using strain gauges. The strain-gauge-based force sensor and its circuitry were fabricated in laboratory following a proved methodology [37] and detailed by a previous graduate student. The assembly, consisting of 4 wire, precision strain gauges, and a half-bridge Wheatstone configuration with amplification and filtering, was inherited as a completed system. In the present work, I performed calibration and validation of the sensor response, and subsequently employed it for torque measurements in the experimental setup.

The following space lists all materials used and figure ref shows the whole experimental setup for a) calibration and b) torque measurement



1. Power supply
2. Ammeter
3. Helmholtz coils
4. Universal support with gaussmeter probe and gaussmeter
5. Circuit including noise calibration
6. DAQ
7. Main device
8. Calibration boat, pipet, and 200 μL tube

Figure 2.7: Experimental equipment and corresponding list.

Helmholtz coils, a pair of PASCO scientific 200 turns coils was used to produce the static uniform magnetic field. This coils were chosen for their ability to increase B in magnitude easily, useful for us to mimic changes of the MRI static magnetic field with distance.

1. Subject (small rod, big rod, lead)
2. Ruler
3. Ceramic bearing
4. Pole
5. Ring
6. Tube
7. Clamp
8. Gauge sensor
9. Computer

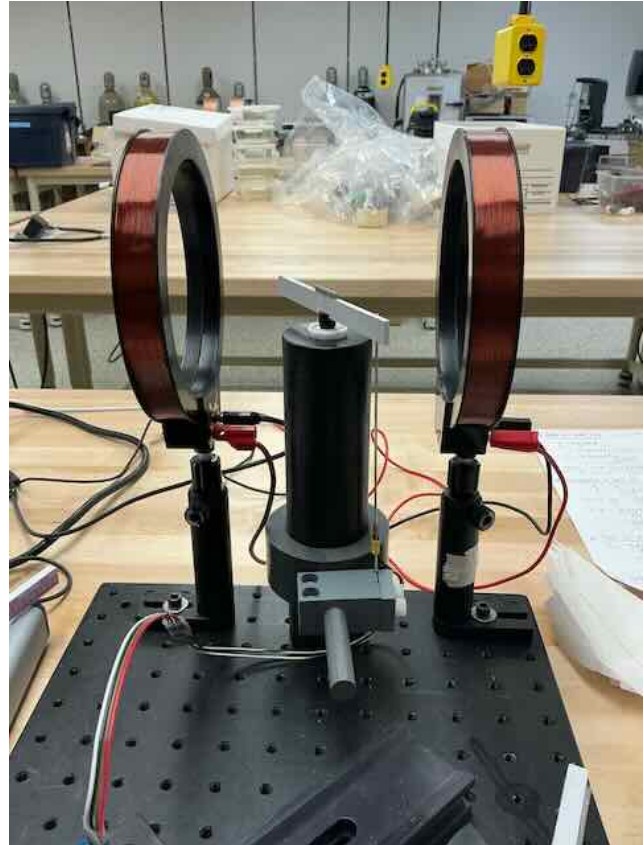


Figure 2.8: Main device close up and corresponding list.

Subject (small rod, big rod, lead): to show the effects of the magnetic field on the setup, a big ferromagnetic rod was positioned on top of the ceramic bearing. It was around 10cm long and with it a strong dipole moment could be induced, serving as a visual confirmation that the system is working.

However a smaller rod is needed, this rod would experiment much smaller torques based on size only. And following the same logic as in translational this would be a better test subject at lab scale. With a small magnetic field, high magnetic susceptibility and small volume, following equation ref proportionality is ensured.

Ruler, the torque is mechanically transmitted from the subject to the ruler and then the contratorque is measured by the sensor. Do I need to explain the physics of this? If so where?

Ceramic bearing (why a ceramic bearing? friction and magnets)

Gauge sensor, set of sensors from different materials were fabricated. These included glass, aluminum and clear plastic. As table 2.1 illustrate, they were made of similar sizes to test repeatability.

Table 2.1: Characteristics of the rods used in experiments.

Rod ID	Material	Diameter (mm)	Length (mm)
Rod1	Al	1.6	145.0
Rod2	Al	1.6	182.00
Rod3	Al	1.6	182.00
Rod5	glass	3.2	166.0
Rod7	clear plastic	2.9	180.0

During calibration, aluminum based sensor was proven to be the best. Comparison plots can be found in appendix B

Circuit including noise calibration, should I include a diagram and explanation of this circuit? If so should I attach it to the appendix as well? DAQ, Computer, a VI was developed for the purpose of registering the voltages a flowchart would be useful here as well.

System calibration

For calibration in the lab, the custom-built sensor was flipped so the bar face horizontally and a small cone-shaped basket made out of tape was attached to hold increments of water. Water was added in μL increments (equivalent to milligrams) simulating torque loads from 0 to 800 mg. Initial increments of 10 up to 200 μL covered the expected force range of epicardial leads weight, followed by 50 μL increments to demonstrate linearity and sensor stability.

From Fig B.6 the smallest measurable force the instrument can be determined. Using the slope of the most reliable fit of Aluminum rod $m = -3.4132e - 05(\leq 0.29\%), \chi_{red} = 1.05$.

The slope relates voltage changes to mass changes:

$$\Delta V = \text{slope} \cdot \Delta m$$

so the smallest measurable mass is

$$m_{min} = \frac{\Delta V}{|\text{slope}|}$$

In this case, the standard error in voltage is $\Delta V = 1.4123e - 06$ which results in $m_{min} \sim 0.0414$ mg. This corresponds to a force of roughly $0.41\mu\text{Newton}$, which is significant because forces expected on an epicardial lead under a MRI magnetic field are extremely small.

Along with these measurements we also tested sensitivity along the length of sensor showing that the further out from the gauge bridge the more sensitive the sensor was. This is demonstrated on an appendix B figure B.7.

In order to find the best position for sensor we set the following experiment, a set of neodymium magnets were aligned to the ceramic bearing to get a static magnetic field, and a platform angled at 45° carried the sensor as we moved it inwards.

From theory, we can show that the maximum torque should appear at this angle - see ch ref - , and then we tested for the optimal length and position from the center, once the torque is transmitted from the metallic rod onto the ruler and it applies the force to the sensor. This process was repeated to find the sweet spot on distance from the center up to a 100 mm radius and the best height.

Figure B.7 shows a small voltage for all distances from 50mm to 100mm and a range of voltages once we are high up (far from the bridge) with higher sensitivity near the center at 50mm.

In Lab Measurement Protocol

These features and measurements lead to a new design of the apparatus. One that is more portable and allows to leverage on the sweet spots of the system. A PVC ring with a small tube for the clamp to sit was added and with it more repeatability was ensured.

During the experiment 2 measurements were performed. Magnetic field flux in Gauss using the **AlphaLab Inc. Gaussmeter**, described before, and Voltage from the strain gauge plus circuit sensor.

The experiment consisted of, with the sensor fixed in the ideal position, vary the magnetic field produced by the Helmholtz coils -mimicking the behavior in distance as a real MRI-increasing as we get closer to isocenter. Then the respective voltage produced by the deformation of the sensor is recorded. Finally, the voltage to weight calibration serves to estimate the induced torque of the system.

In Clinic Measurement Protocol

Torque measurements were conducted ensuring consistency and reproducibility. Prior to setup, both researchers confirmed that all personal belongings and ferromagnetic items had been removed from their pockets. The VI was opened with the necessary parameters configured, and the data acquisition card (DAC) was powered on. A 10-meter jumper wire was connected to the DAC and integrated into the circuit, with the computer system interfaced to the DAC for real-time data collection.

The experimental apparatus was then fitted with the test object—either the magnetically susceptible calibration material or the epicardial lead—mounted on the rotating ruler. Then it was positioned on the MRI room bench at a distance of 1 meter from the bore, measured from the base of the apparatus to the bore face. Finally, the ruler arm was aligned to rest against the strain gauge, oriented as close as possible to 45° relative to the bore axis.

In the control room, torque measurements were acquired according to a predefined flowchart procedure, with results recorded on the MEASUREMENT WORKSHEET. A section of the worksheet was considered complete only when all data fields were marked with an “X.” Section 1 of the worksheet corresponded to measurements performed with the magnetically susceptible calibration object. Following these measurements, the apparatus was retrieved from the MRI room, and the calibration object was replaced with the epicardial lead affixed to the ruler. Section 2 of the worksheet was then completed by repeating the flowchart process with the lead in place.

Data collection in each section was performed at incremental positions along the MRI bore axis. For both the calibration object (Section 1) and the epicardial lead (Section 2), torque measurements were acquired at eleven translational positions: 100, 90, 80, 70, 60, 50, 40, 30, 20, 10, and 0 cm from the bore face. This systematic sampling allowed for direct comparison of torque response between the calibration object and the epicardial lead across the same magnetic field gradient.

Chapter 3

Results

overall description of the experiments and presentation of the data for the following:

3.1 Magnetic Field Mapping

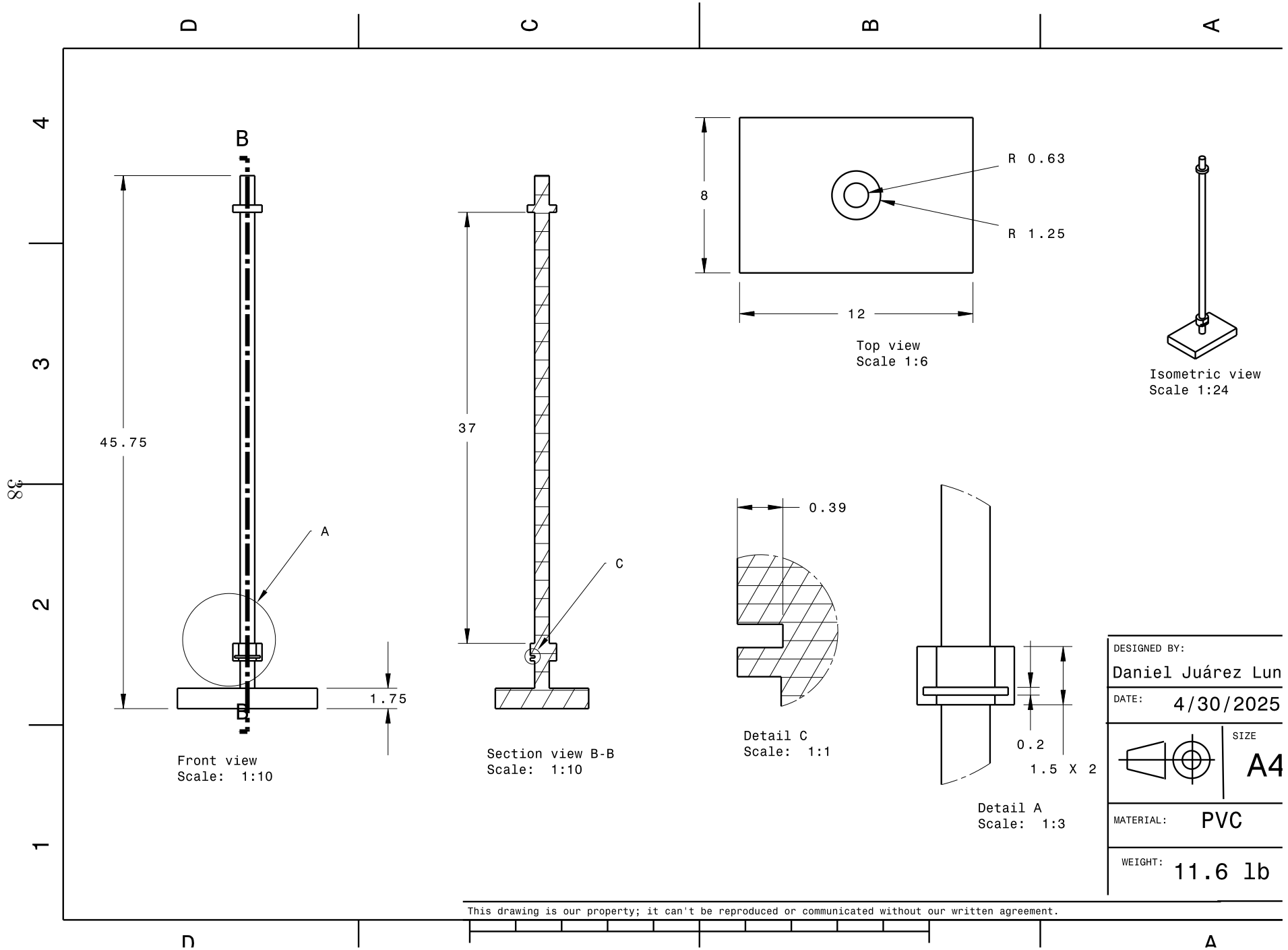
3.2 Translational

3.3 Torque

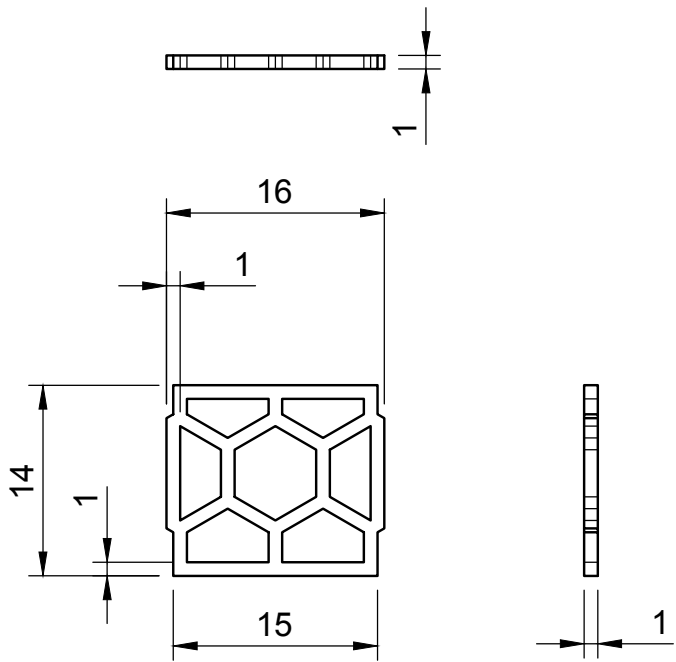
Appendix A

Engineering Drawings

Appendix A1: Pole drawing



Properties
Material: PLA
Weight:0.124 g



Dept.	Technical reference	Created by Daniel Juarez	8/5/25	Approved by
		Document type	Document status	
		Title Lead Holder Bee Comb Mesh MRI Translational Force	DWG No.	
		Rev.	Date of issue	Sheet 1/1

Appendix B

Calibration Logs

Additional experimental relationships and fitting diagnostics are presented to support the main analysis.

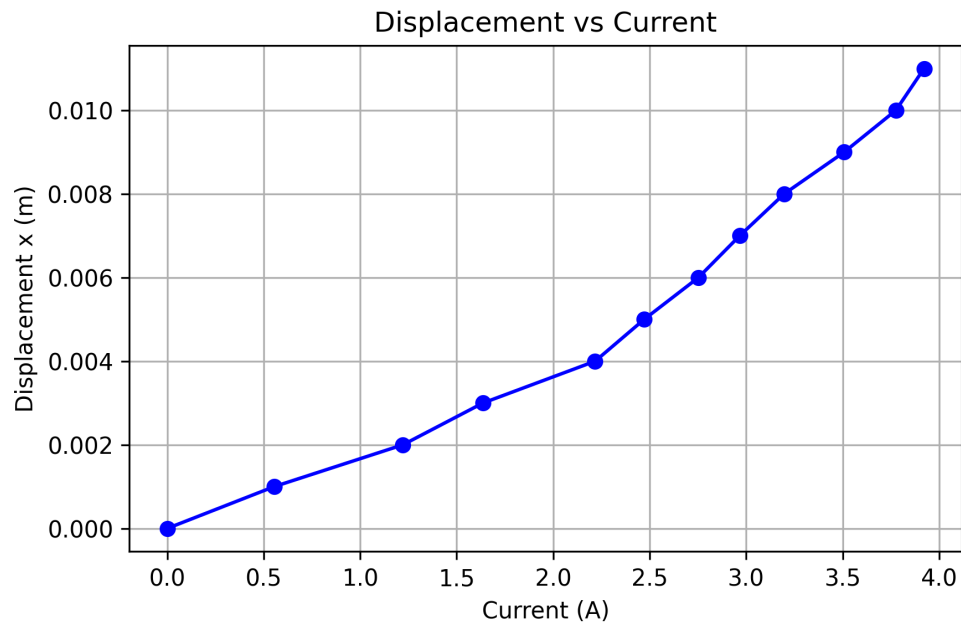


Figure B.1: Displacement as a function of current through the solenoid.

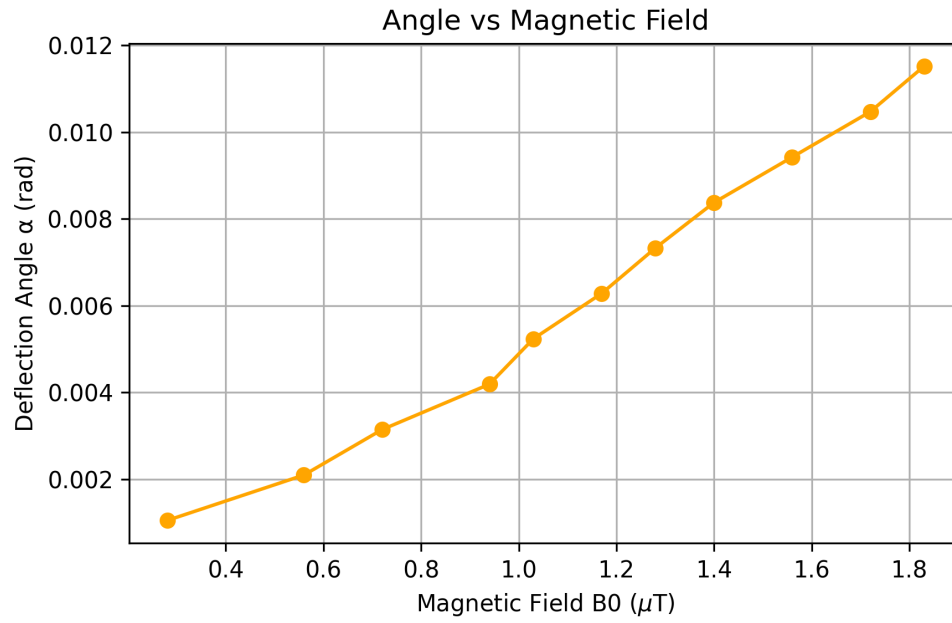


Figure B.2: Measured angle α versus magnetic field intensity B_0 .

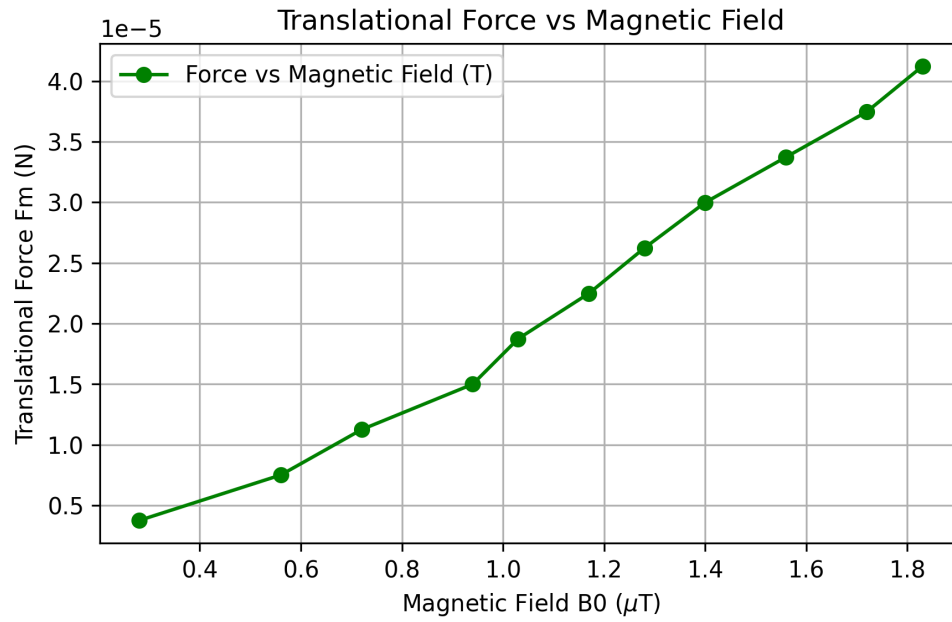


Figure B.3: Computed translational force F_m as a function of B_0 .

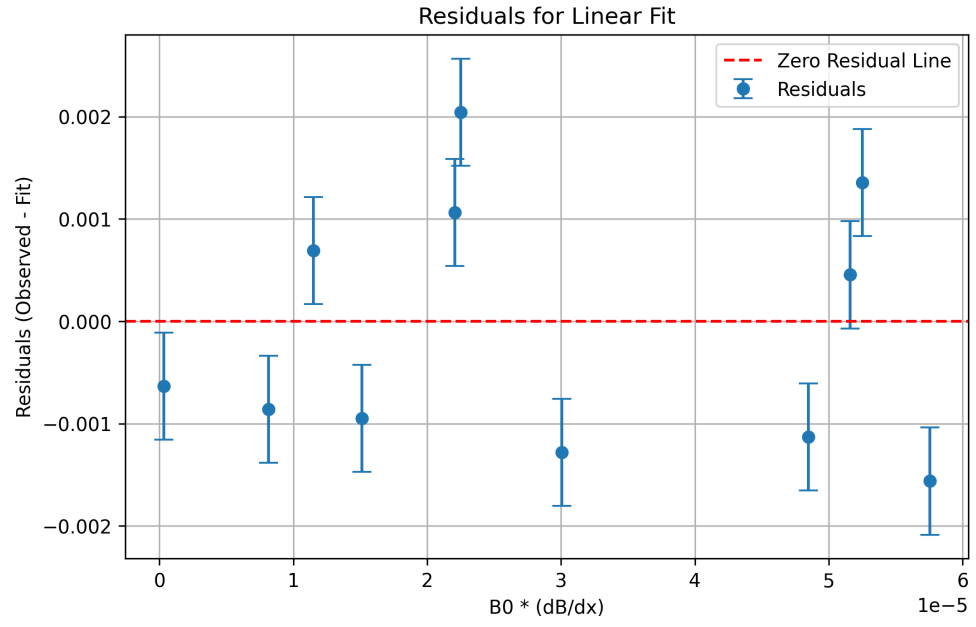


Figure B.4: Residuals from linear fit of $\tan(\alpha)$ vs. $B_0 \cdot \frac{dB_0}{dz}$. Error bars indicate propagated uncertainties.

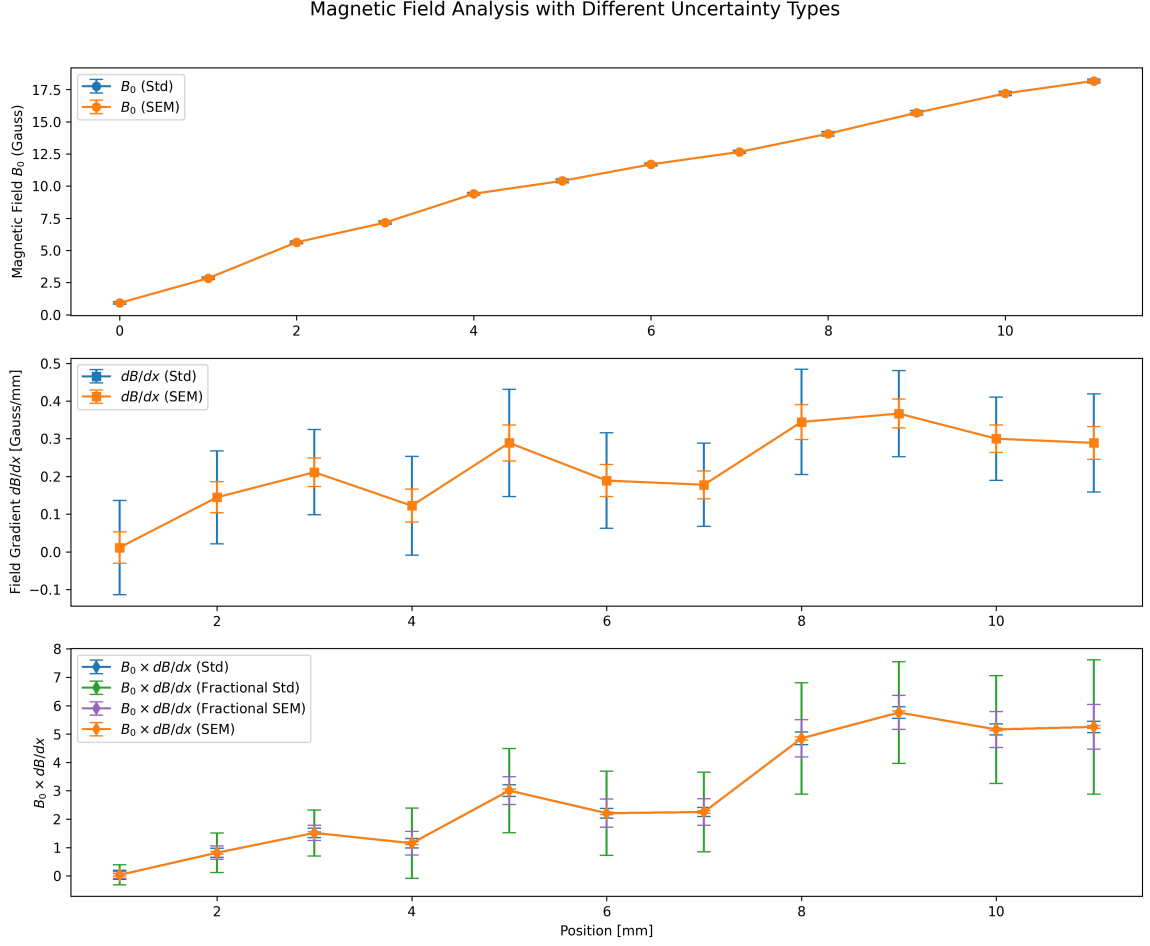


Figure B.5: Propagation of measurement uncertainty through magnetic field analysis. Top: Measured field B_0 with standard deviation and SEM. Middle: Gradient dB/dz with propagated error. Bottom: Final product $B_0 \cdot dB/dz$ with multiple uncertainty estimations (standard, fractional, SEM). This term was used as the independent variable in the susceptibility fit.

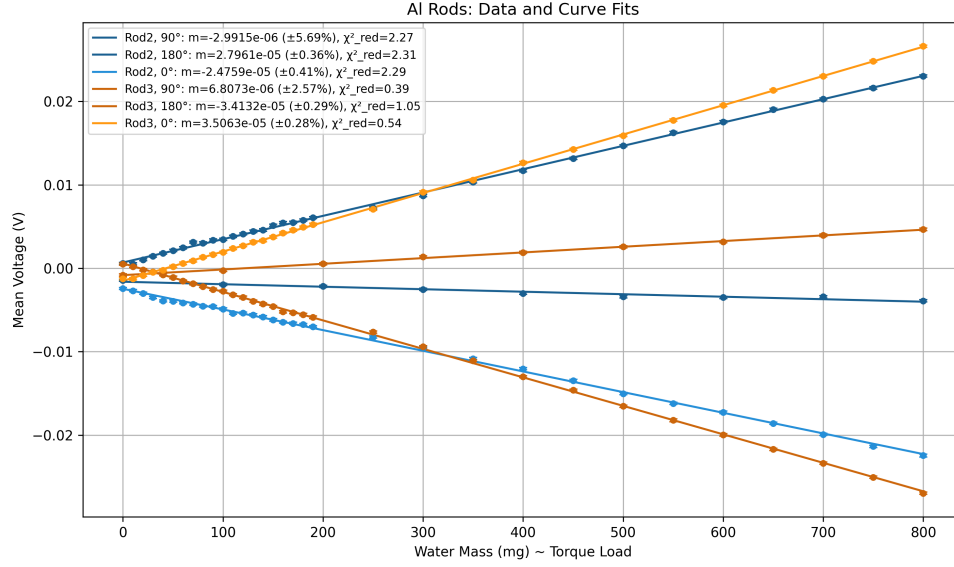


Figure B.6: Calibration curves for two aluminum (AL) rods in three different orientations. Linear fits show consistent response across orientations, confirming the reliability of the lab-built sensor (see Methods Section for details).

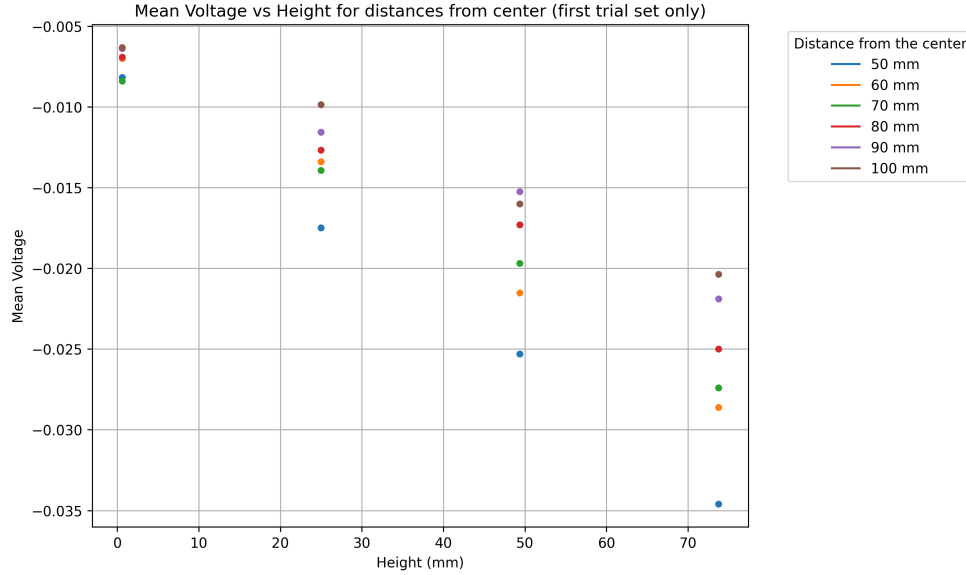


Figure B.7: Mean voltage response of the sensor as a function of height for different distances from the center (50–100 mm). A small and relatively flat voltage is observed across all radii, while at larger heights (farther from the bridge) the measurements diverge, showing higher sensitivity near the center (50 mm). See Methods Section for details on torque transfer and distance selection.

Appendix C

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Glossary

CDR Creighton Digital Repository. 45

CIED cardiac implantable electronic device. 1, 2, 9

CRT cardiac resynchronization therapy. 9

DAC data acquisition card. 34

ECG electro cardigram. 1

HRS Heart Rhythm Society. 1

ICD implantable cardioverter-defibrillators. 9

MRI Magnetic Resonance Imaging. 1, 2, 3, 4, 6, 14, 19, 20, 25, 30, 33

PD Proton Density. 4

PLA Polylactic Acid. 24

PVC Polyvinyl Chloride. 15, 19, 22, 29, 34

RF radiofrequency. 1, 2, 3, 4, 14

VI LabVIEW Virtual Instrument (VI). 20, 21, 32, 34

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